# CycleTwin

Cycle-by-cycle late or absent ovulation research

Team CycleTwin / VIT

A fitted synthetic forecasting workflow with explicit safety gates

Meet Kushi: fictional persona, never performance evidence

Synthetic research prototype. Not for diagnosis, contraception or conception planning.

# The research problem

Monitoring irregular cycles without claiming a PMOS diagnosis

Target: generated ovulation after the configured cutoff, or generated absence

India context: 7.2% by NIH criteria and 19.6% by Rotterdam criteria

Published prevalence depends on definition. It is not the simulator mixture or model accuracy.

# Meet Kushi

Fictional synthetic persona - not a real patient - not performance evidence.

21 years, year 3 B.Tech

Hostel life, exam-season sleep debt, mother with type 2 diabetes, smartwatch wearer.

Actual generated past lengths: 41, 38, 41, 38, 44, 51 days

Selection uses generator truth and a declared seed, never predicted risk

# Static and dynamic input streams

EHR-lite is implemented and explicitly oracle-adjacent

Static: generated IR latent, androgen, age, BMI and family history

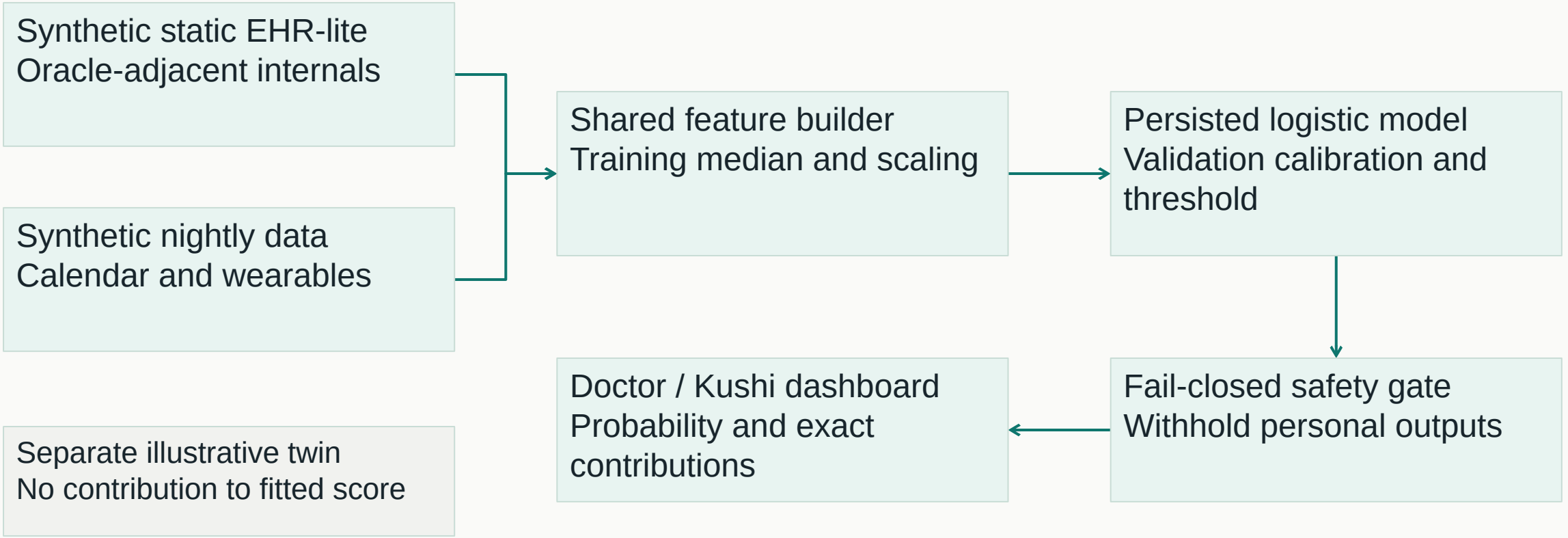
Dynamic: calendar history, RHR, HRV, steps, sleep and temperature

No measured hormone values. The XML parser demonstrates input shape only.

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# One fitted prediction path

The dashboard loads the same persisted pipeline used by the benchmark




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# Training and selection

Every explicit operating choice uses train or validation data

Patient counts per development seed: train 840, validation 180, test 180

Training medians and scaling, logistic regression, validation Platt calibration

Validation Brier-gain rule selects richer features or calendar fallback

Validation F1 chooses the threshold. Test outcomes never choose that threshold.

# Synthetic development evidence: day 10

Seed means describe the current generator, not clinical forecasting

| Day | Tier | Mean AUROC | Mean Brier | Mean day MAE |
|---|---|---|---|---|
| 10 | 0 | 0.838 | 0.163 | 5.004 |
| 10 | 1 | 0.838 | 0.163 | 5.004 |
| 10 | 2 | 0.838 | 0.163 | 5.004 |
| 10 | 3 | 0.838 | 0.163 | 5.004 |

# Synthetic development evidence: day 14

Seed means describe the current generator, not clinical forecasting

| Day | Tier | Mean AUROC | Mean Brier | Mean day MAE |
|---|---|---|---|---|
| 14 | 0 | 0.838 | 0.163 | 5.132 |
| 14 | 1 | 0.838 | 0.163 | 5.132 |
| 14 | 2 | 0.850 | 0.157 | 6.069 |
| 14 | 3 | 0.843 | 0.161 | 5.565 |

# Synthetic development evidence: day 18

Seed means describe the current generator, not clinical forecasting

| Day | Tier | Mean AUROC | Mean Brier | Mean day MAE |
|---|---|---|---|---|
| 18 | 0 | 0.838 | 0.163 | 5.439 |
| 18 | 1 | 0.839 | 0.163 | 5.439 |
| 18 | 2 | 0.906 | 0.119 | 5.267 |
| 18 | 3 | 0.907 | 0.119 | 5.267 |

# Frozen synthetic evidence: day 10

Reserved seeds, unchanged train/validation choices, one guarded run

| Day | Tier | Mean AUROC | Mean Brier | Mean day MAE |
|---|---|---|---|---|
| 10 | 0 | 0.839 | 0.164 | 5.142 |
| 10 | 1 | 0.839 | 0.164 | 5.142 |
| 10 | 2 | 0.839 | 0.164 | 5.142 |
| 10 | 3 | 0.839 | 0.164 | 5.142 |

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# Frozen synthetic evidence: day 14

Reserved seeds, unchanged train/validation choices, one guarded run

| Day | Tier | Mean AUROC | Mean Brier | Mean day MAE |
|---|---|---|---|---|
| 14 | 0 | 0.839 | 0.164 | 5.251 |
| 14 | 1 | 0.839 | 0.164 | 5.251 |
| 14 | 2 | 0.855 | 0.155 | 6.790 |
| 14 | 3 | 0.850 | 0.158 | 6.319 |

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# Frozen synthetic evidence: day 18

Reserved seeds, unchanged train/validation choices, one guarded run

| Day | Tier | Mean AUROC | Mean Brier | Mean day MAE |
|---|---|---|---|---|
| 18 | 0 | 0.839 | 0.164 | 5.570 |
| 18 | 1 | 0.839 | 0.164 | 5.570 |
| 18 | 2 | 0.907 | 0.117 | 5.449 |
| 18 | 3 | 0.907 | 0.117 | 5.449 |

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# Evidence at the latest sensor snapshot

Predeclared temperature sensor tier, descriptive seed means

Synthetic day 18, Tier 2: AUROC 0.907, Brier 0.117

3,240 synthetic test cycles across 3 frozen seeds

A separate frozen evaluation artifact is available.

Standard average precision and calibration diagnostics are available from stored predictions.

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# What the clinician view shows

Decision-support research display

Fitted probability beside its own validation threshold and held-out model evidence

Exact additive feature contributions and explicit imputation

Heuristic day estimate and conditional, illustrative day band

Data-quality ribbon and the reason for any withholding

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# Guardrails live

Missing safety information causes abstention

Contraception, thyroid and prolactin flags, age and pregnancy-related states

Early post-menarche, perimenopause and insufficient observed temperature

Withholding removes personal scores, intervals, explanations, sandbox outputs and alerts

The workflow is synthetic and its safety effectiveness has not been clinically evaluated

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# The twin and assumption sandbox

Clear boundaries around simulated physiology

Reproductive and metabolic equations generate an illustrative state trajectory

The twin does not affect the logistic score or its reported performance

The sandbox edits invented model inputs. It does not predict a treatment effect

Glucose threshold crossings are simulated counts, not an evaluated glucose forecast

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# Where it fails

All declared Kushi scenarios remain visible

Display the fixed scenario seeds with misses, false alarms or withholding unchanged

Different snapshots can fall back to calendar even when sensors are available

A class set containing both outcomes does not identify one result

Robustness, state-estimator and conformal coverage evidence remain separate work

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# Validity limits

Synthetic recovery is not prospective evidence in women

Labels and sensor changes share generator assumptions

EHR-lite includes a label-generating internal latent

Old test seeds were reused during development

The current benchmark does not evaluate the complete sequential alert policy

# Reproducible demonstration

Public source, inspectable artifacts and explicit next work

README, model cards and the audit link every claim to evidence

Persisted inference reproduces the benchmark and exposes exact feature calculations

The required video remains a human recording and upload task

Clinical validation, clinician feedback and sim-to-real evaluation remain outstanding

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